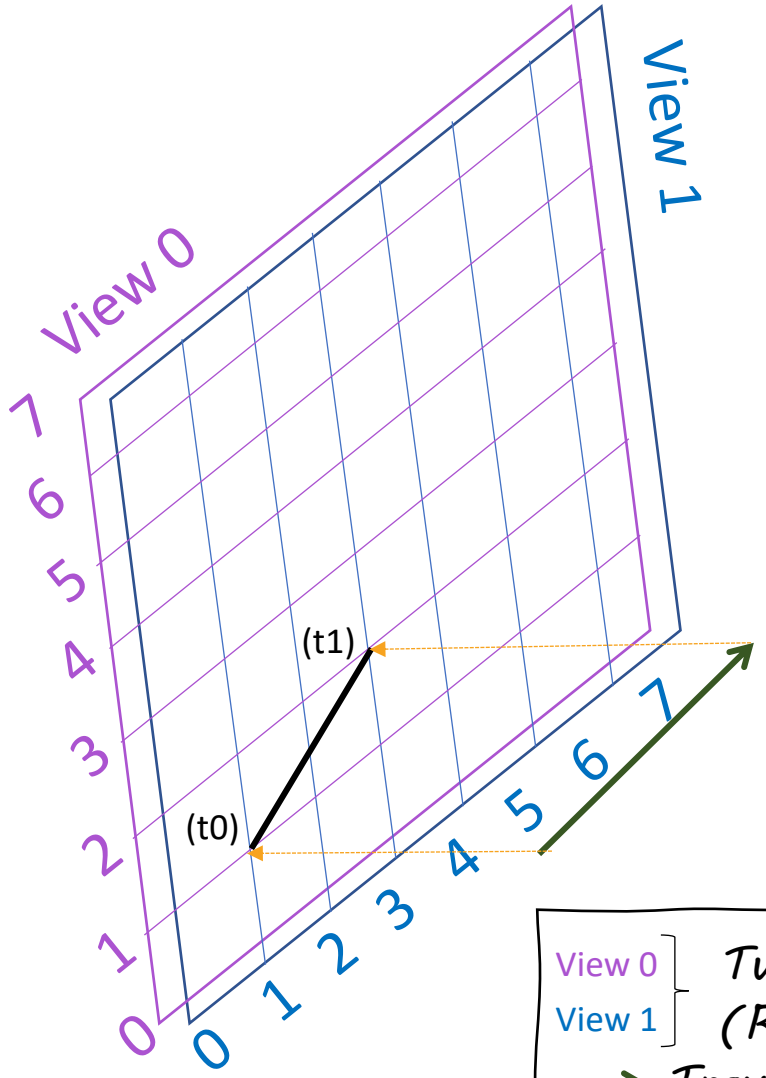
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LArTPCs images: a step-by-step example



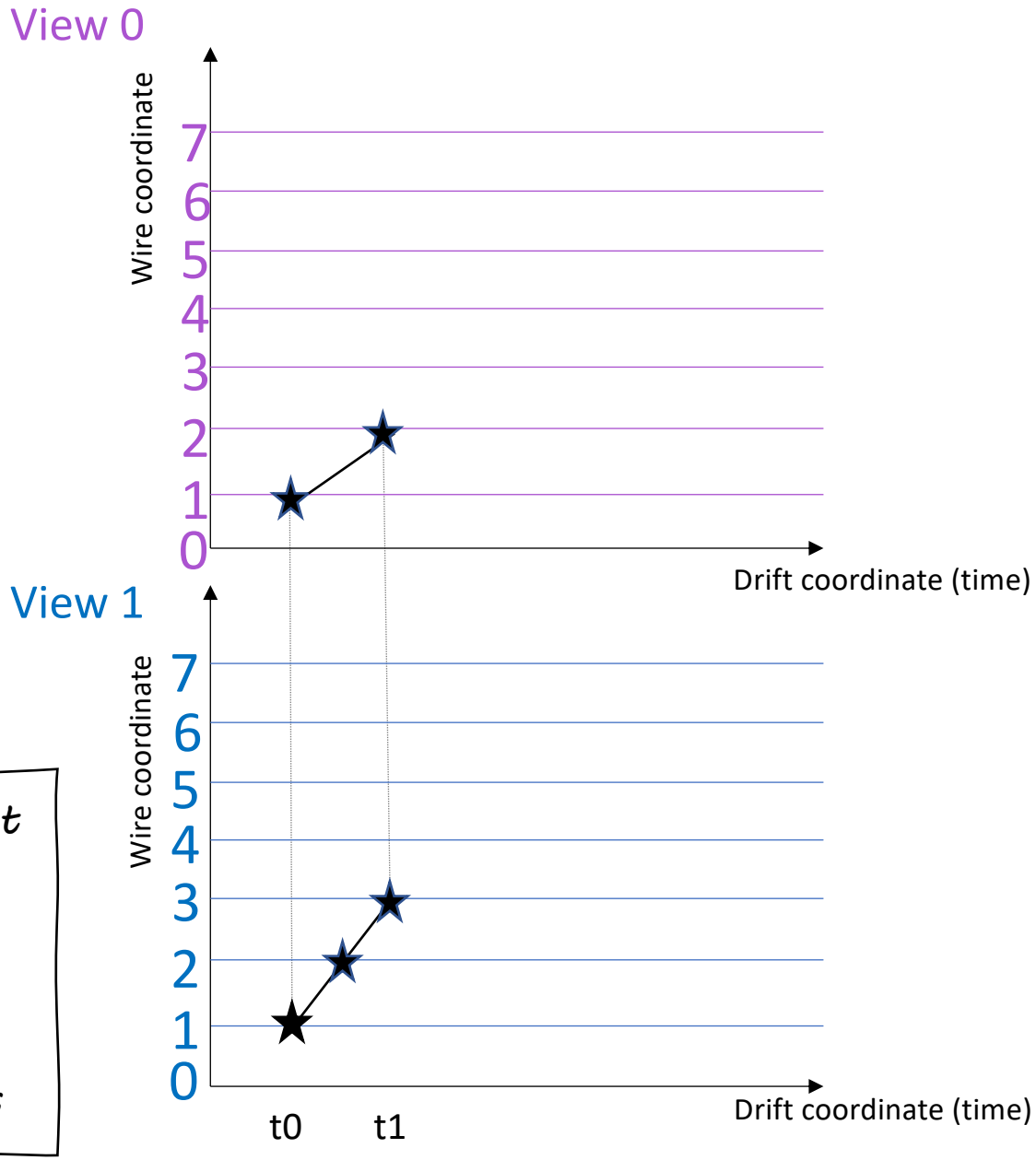
View 0 — Two LArTPC readout (RO) planes
View 1

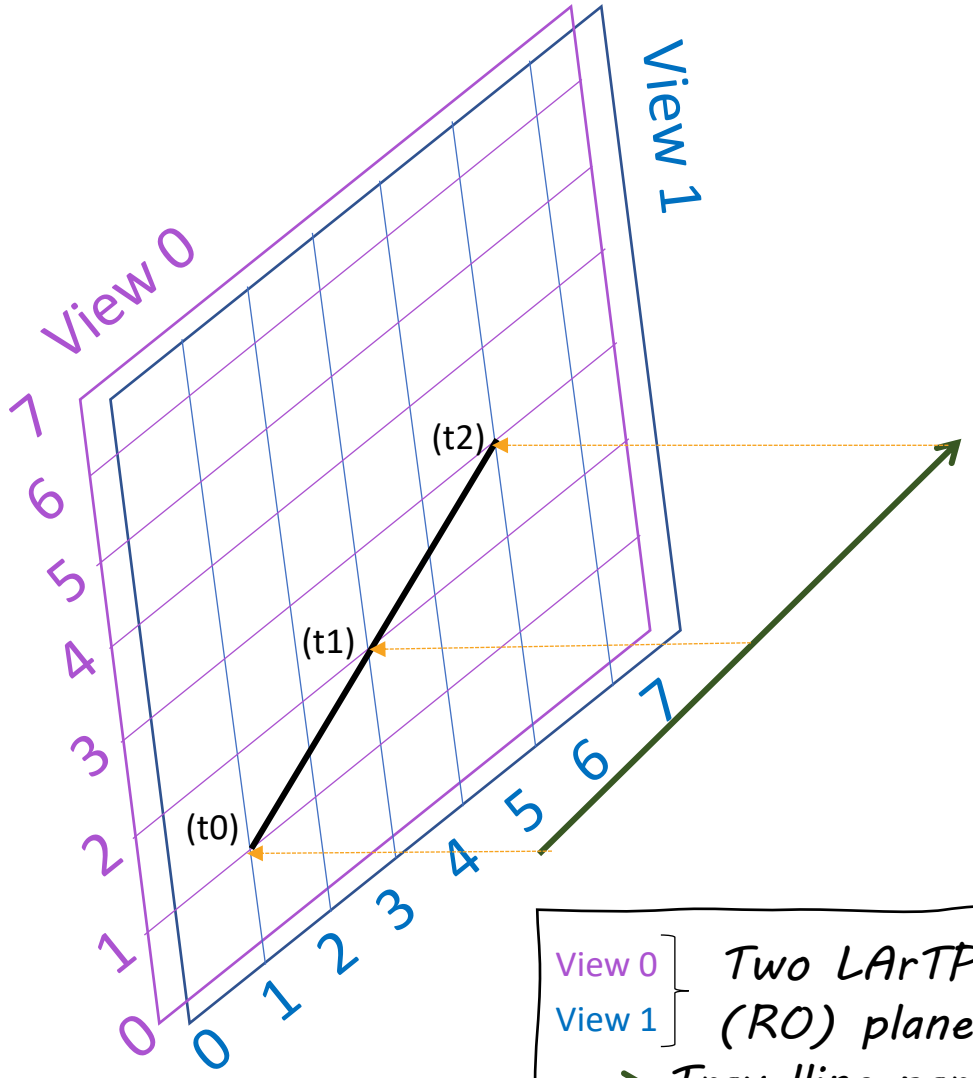
Travelling particle

Ionisation electrons

Electron arrival positions on RO planes

Electron drift time (~ms)





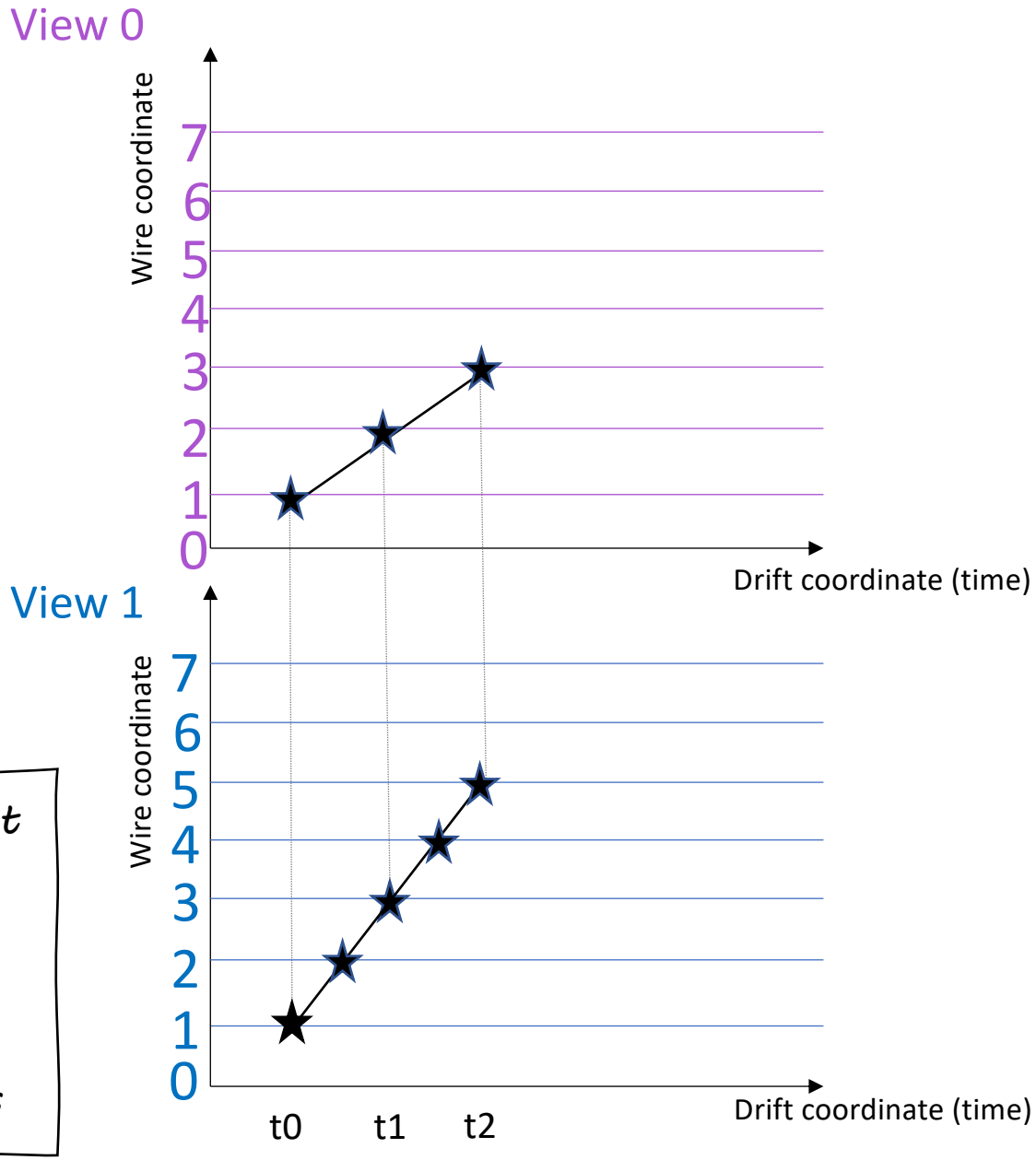
View 0 / View 1 Two LArTPC readout (RO) planes

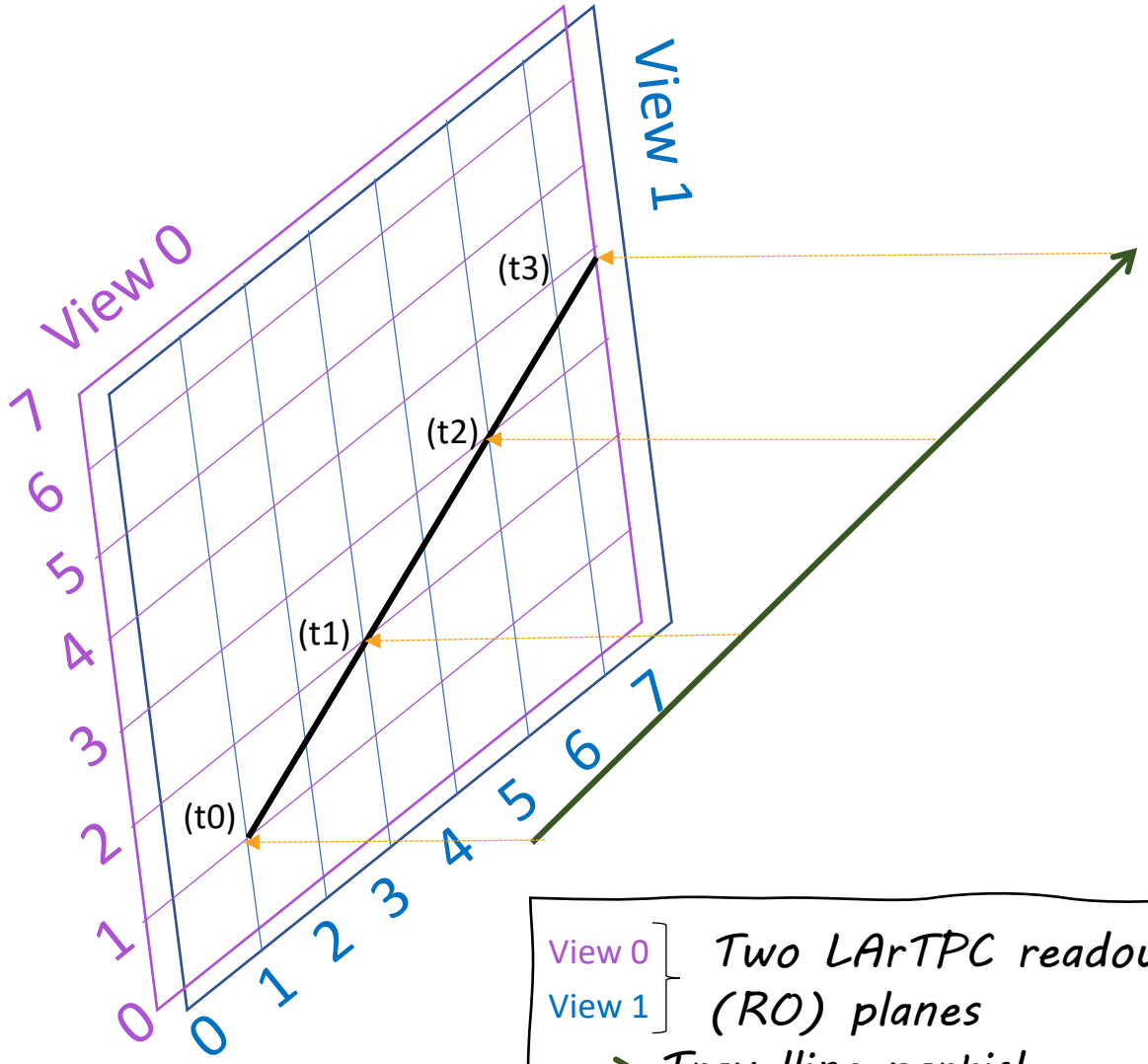
Travelling particle

Ionisation electrons

Electron arrival positions on RO planes

Electron drift time (~ms)





View 0 — Two LArTPC readout (RO) planes

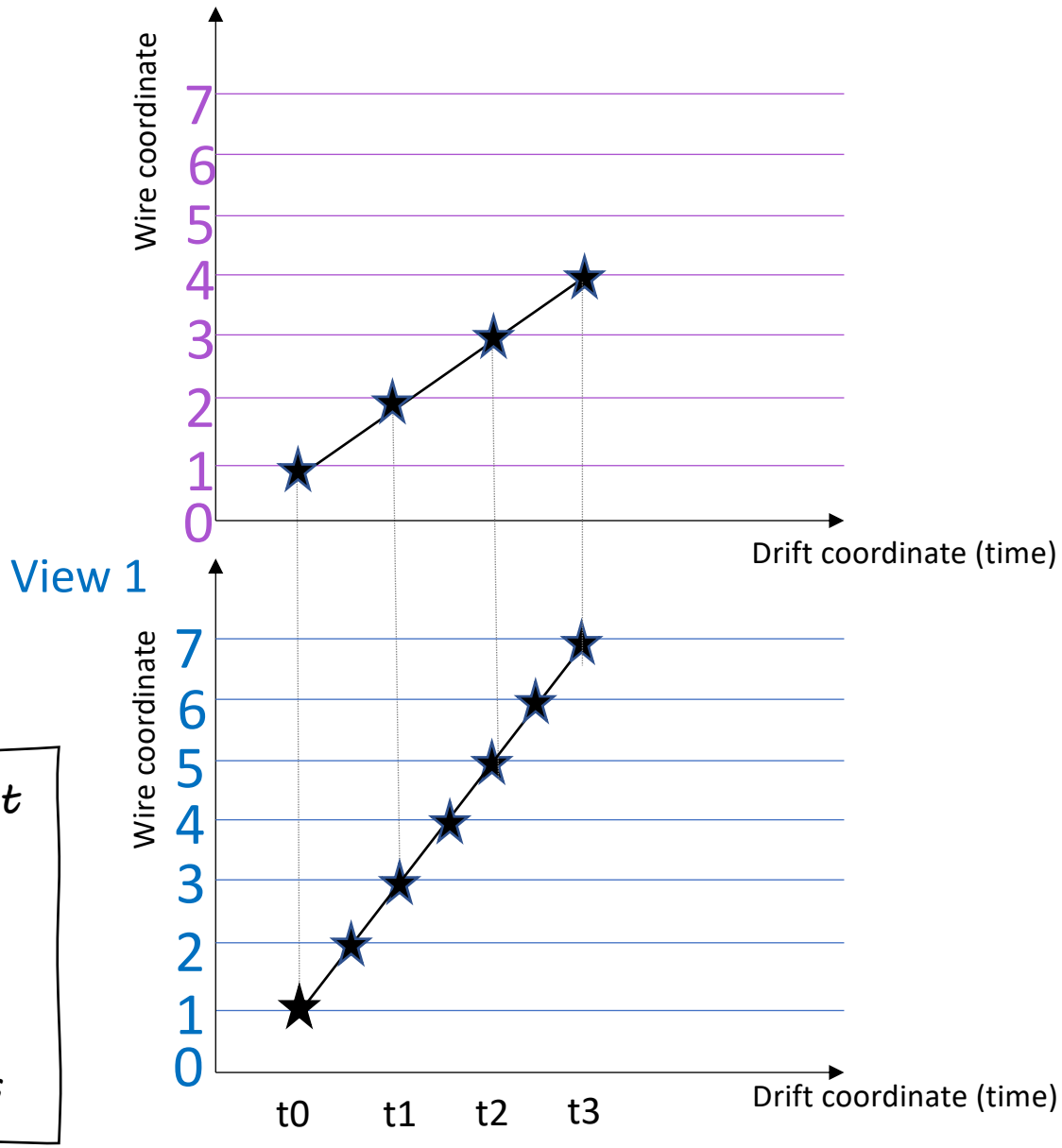
View 1

Travelling particle

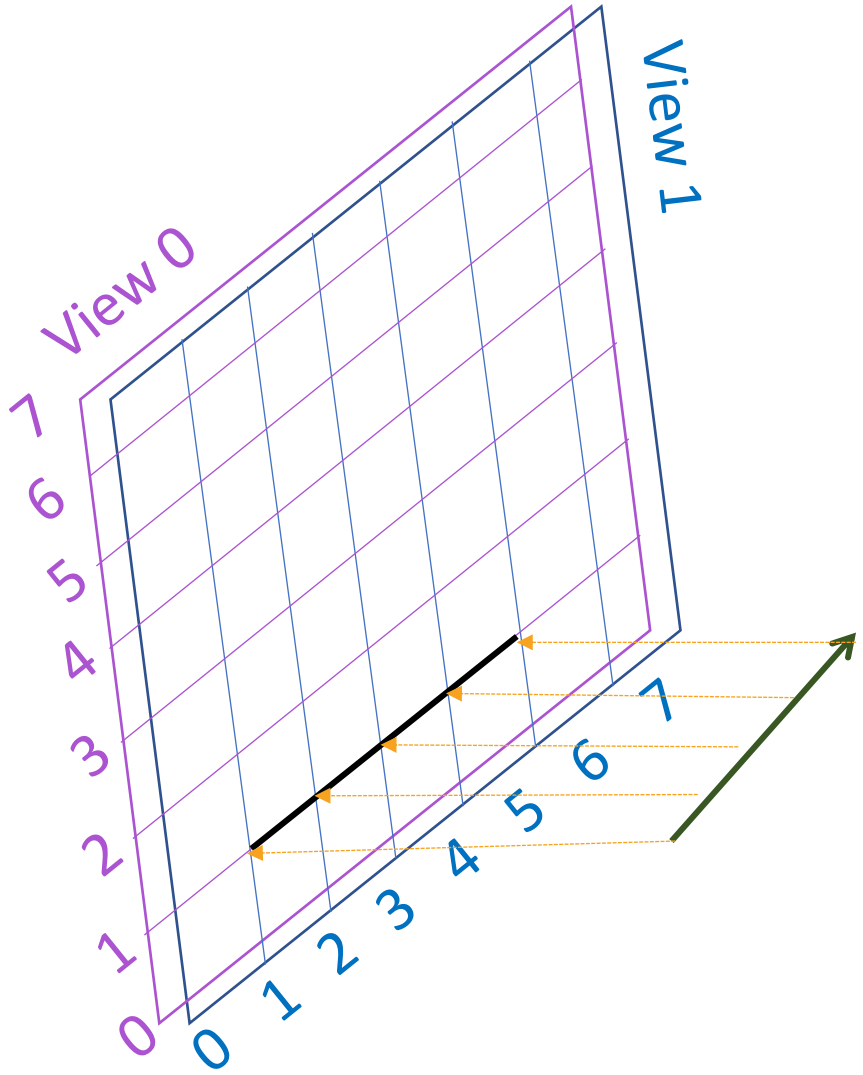
Ionisation electrons

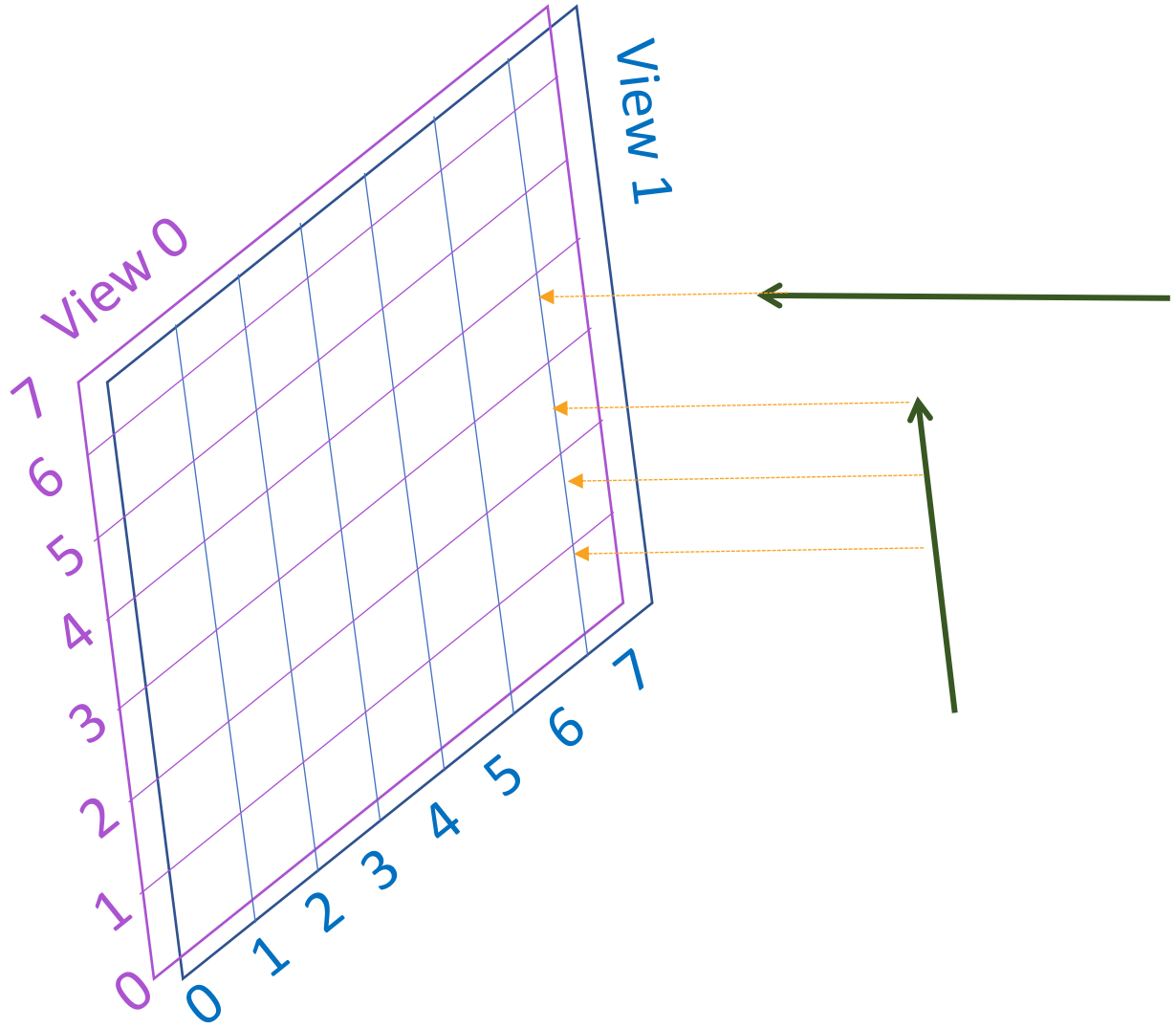
Electron arrival positions on RO planes

Electron drift time (~ms)



Electron drift time (~ms)





What happens in these cases?

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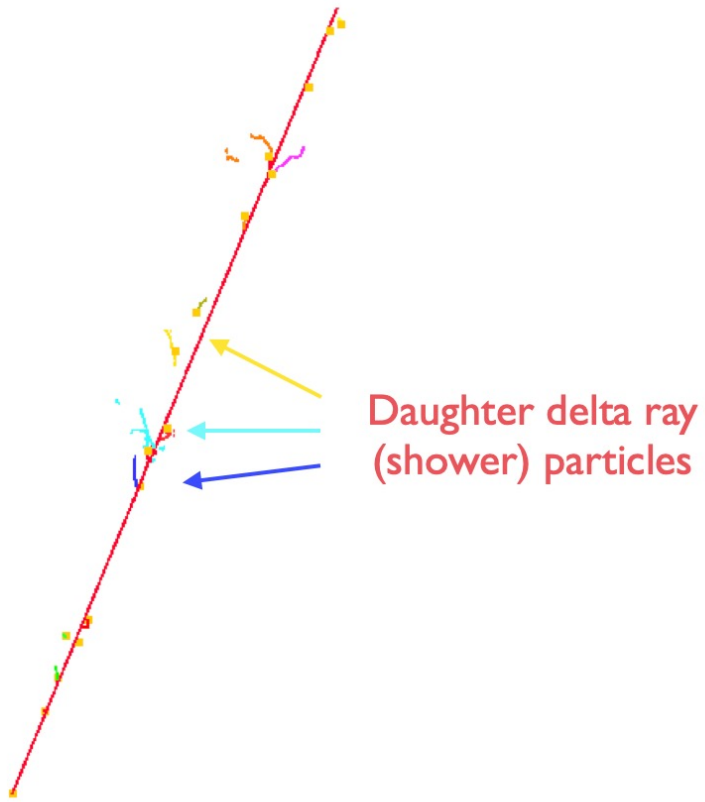
Main particle topologies in LArTPCs

Muons in LArTPCs

- Produce track-like topologies in LArTPCs

- Make ***delta rays*** as they travel by scattering with electrons at rest

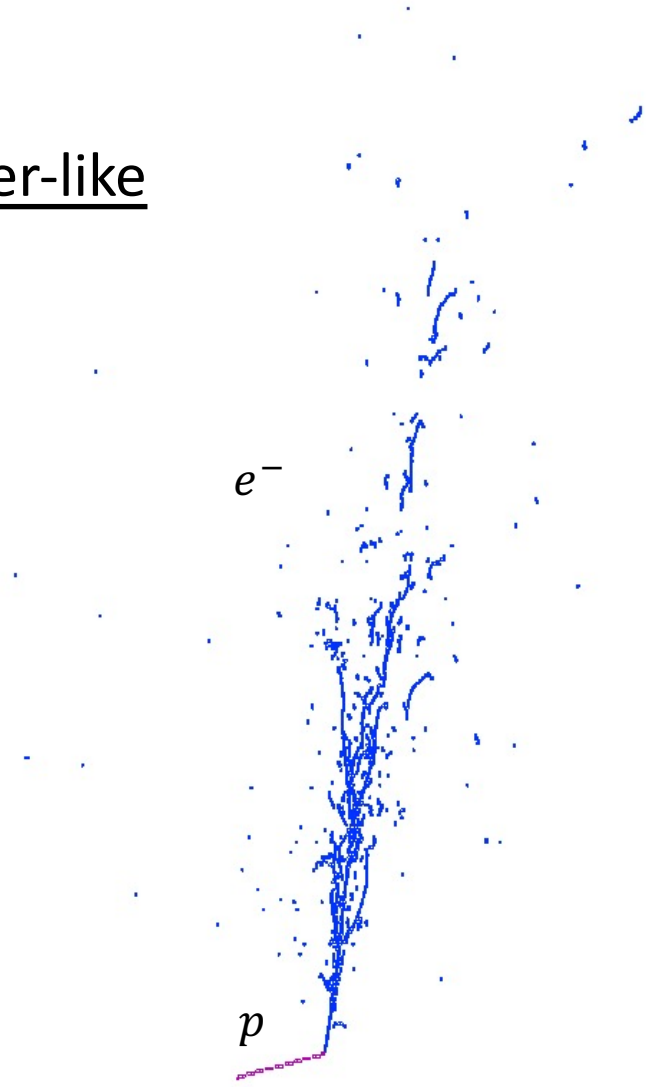
- Muon decay: $\boldsymbol{\mu^- \rightarrow e^- \overline{\nu}_e \nu_\mu}$



Electrons and photons in LArTPCs

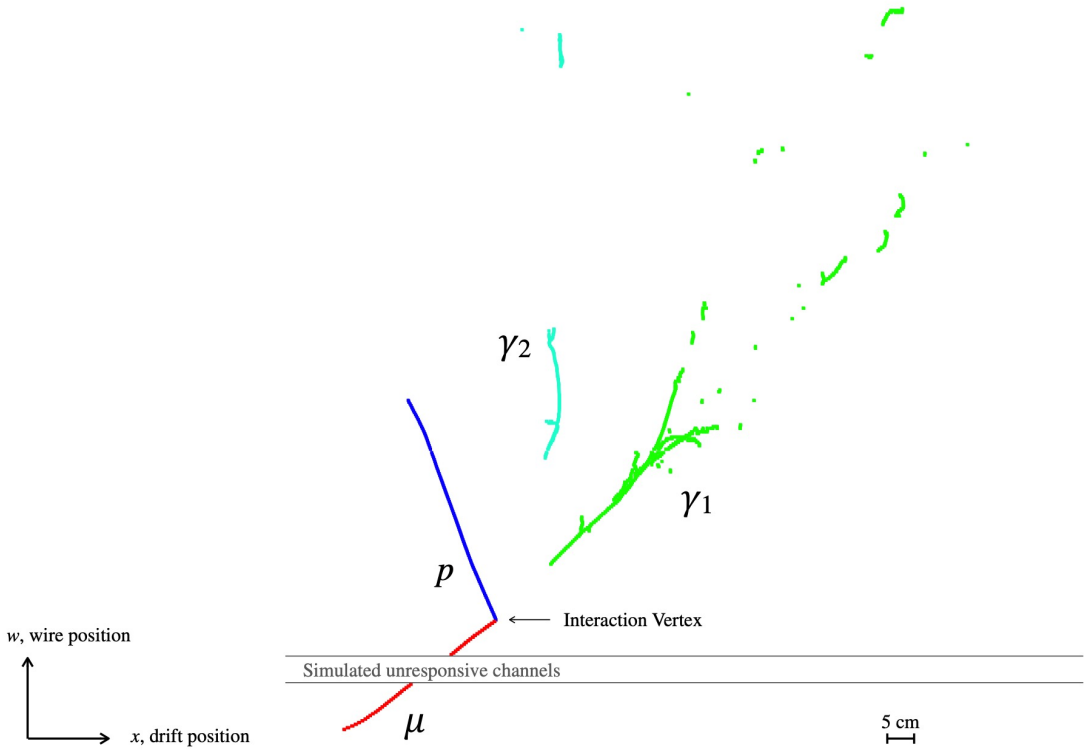
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- Produce shower-like topologies



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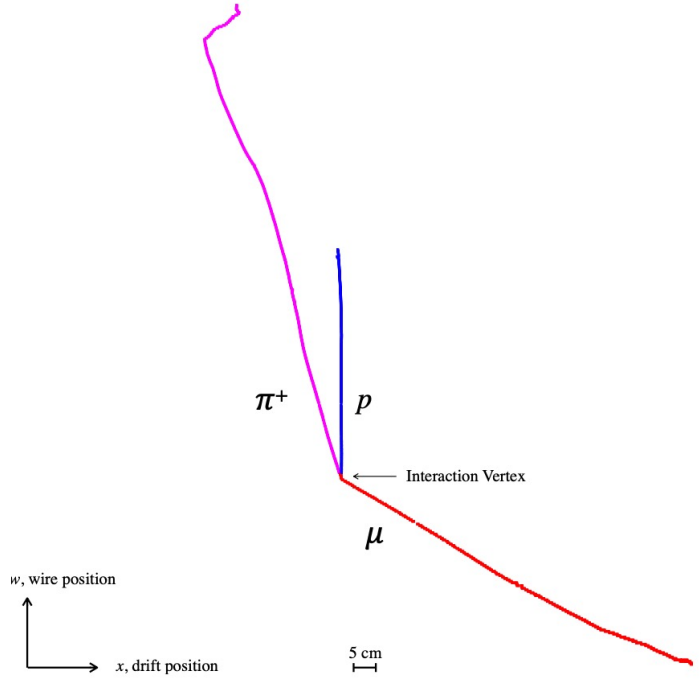
- Can distinguish them via energy profile and "conversion gap"

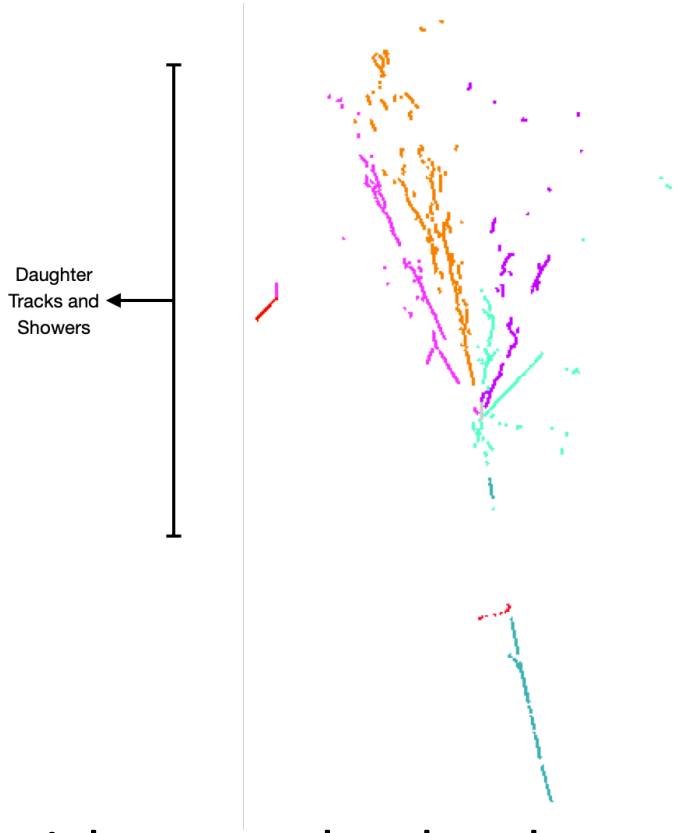


Charged pions in LArTPCs

- Produce track-like and shower-like topologies in LArTPCs
- Decay to muons which in turn decay to electrons

$$\pi^- \rightarrow \mu^- + \bar{\upsilon}_\mu \qquad \mu^- \rightarrow e^- + \bar{\upsilon}_e + \upsilon_\mu$$





Test beam particle example: the charged pion interacts and breaks up an argon nucleus

Protons in LArTPCs

- Typically track-like topologies in LArTPCs
- Higher energy loss → tracks typically short
- Shower-like topologies at higher energy

